1 DH description

The DH frames are assigned from the CuRo6R URDF, with frame $\{0\}$ at `base_link`, joint i rotating about z_{i-1}, and $q = 0$ matching the URDF home configuration. We use standard DH transforms:

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i). \tag{1}$$

Table 1 gives the resulting parameters and the geometric choices that determine them. The only fixed joint-angle offset is $\theta_3 - q_3 = \pi/2$.

Table 1: DH parameters of CuRo6R. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric origin of the parameters
1	1	q_1	$-\pi/3$	0.8	The two base heights add: $d_1 = 0.5 + 0.5$; the lateral offset is $a_1 = 0.8$. Joint 2's axis is its local y axis, so the 30° URDF roll gives $\alpha_1 = 30° - 90° = -60°$.
2	1.25	q_2	$+\pi/2$	2	The joint-3 offset resolves into 1.25 along z_1 and 2 along x_2. At home $x_2 = x_1$, and the orthogonal joint axes give $\alpha_2 = +90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Moving 0.75 along z_2 reaches the intersection with joint 4's axis: O_3 in Fig. 1(a), hence $a_3 = 0$. Choosing $x_3 = y_2$ gives the +90° angle offset and $\alpha_3 = +90°$.
4	2.5	q_4	$-\pi/2$	0	The wrist centre lies $1.5 + 1 = 2.5$ along z_3 from O_3. Axes 4 and 5 intersect there; with $x_4 = x_3$ at home, $\alpha_4 = -90°$.
5	0	q_5	$+\pi/2$	0	Axes 5 and 6 meet at the wrist centre, so $O_5 = O_4$ and $d_5 = a_5 = 0$ [Fig. 1(b)]. Keeping $x_5 = x_4$ gives $\alpha_5 = +90°$.
6	1.5	q_6	0	0	The endpoint is 1.5 along z_5 from the wrist centre. Choosing $z_6 = z_5$ and $x_6 = x_5$ at home gives $a_6 = \alpha_6 = 0$.

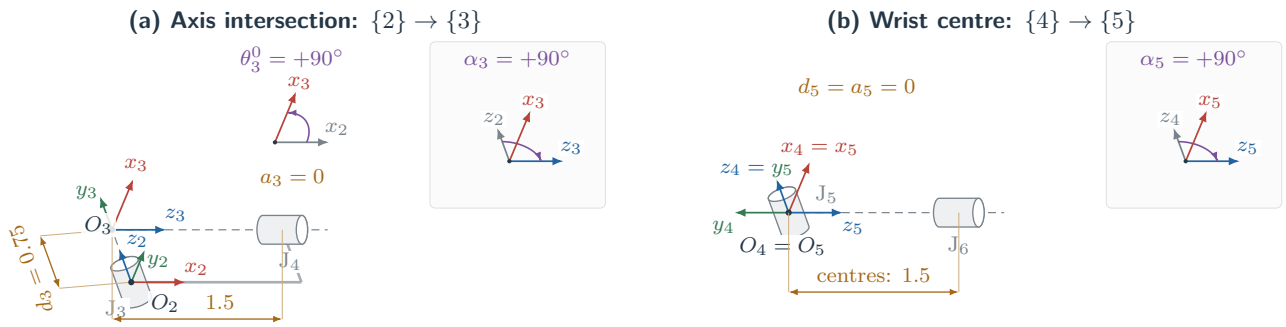


Figure 1: Frame placement at $q = 0$. (a) O_3 lies 1.5 m before the physical origin of joint 4 along its axis; this distance contributes to d_4. (b) $O_4 = O_5$ at the wrist centre; the 1.5 m between joints 5 and 6 contributes to d_6, while $d_5 = 0$. Cylinders mark physical joint origins and black dots mark DH origins; x, y, and z identify the frame axes.

Endpoint alignment. The endpoint E is `link_6`. Its origin coincides with O_6, while $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. The fixed alignment is therefore appended to the DH chain to recover the URDF endpoint pose:

$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

2 Aalytical Inverse Kinematics

Derivation:

The wrist position can be easily found from the desiired end effector pose $T_d = \begin{pmatrix} R_d & p_d \\ 0^T & 1 \end{pmatrix}$:

$$p_w = p_d - d_6 R_d e_x = \begin{pmatrix} x_w \\ y_w \\ z_w \end{pmatrix}$$

To make computations easier, we can undo the waist roation with $R_z(-q_1) = \begin{pmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$:

$$u = R_z(-q_1)p_w = \begin{pmatrix} c_1 x_w + s_1 y_w \\ -s_1 x_w + c_1 y_w \\ z_w \end{pmatrix}$$

In the waist adjusted coordinates, the shoulder (joint 2) is at $s = \begin{pmatrix} a_1 \\ 0 \\ d_1 \end{pmatrix}$. The vector from the shoulder to the wrist center in the waist adjusted coordinates is $v = u - s = \begin{pmatrix} c_1 x_w + s_1 y_w - a_1 \\ -s_1 x_w + c_1 y_w \\ z_w - d_1 \end{pmatrix}$. To make the math easier, we can also undo the shoulder tilt. Let

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = R_x\left(-\frac{\pi}{6}\right) v = \begin{pmatrix} c_1 x_w + s_1 y_w - a_1 \\ -\frac{\sqrt{3}}{2} s_1 x_w + \frac{\sqrt{3}}{2} c_1 y_w + \frac{1}{2} z_w - \frac{1}{2} d_1 \\ \frac{1}{2} s_1 x_w - \frac{\sqrt{3}}{2} c_1 y_w + \frac{\sqrt{3}}{2} z_w - \frac{\sqrt{3}}{2} d_1 \end{pmatrix}$$

In the elbow aligned frame, the vector from the elbow to the wrist center is $b_{\text{aligned}} = \begin{pmatrix} d_4 \\ 0 \\ d_3 \end{pmatrix}$

To express the same vector in the shoulder-aligned frame, we add the elbow rotation: $b = R_z(q_3) b_{\text{aligned}} = \begin{pmatrix} c_3 d_4 \\ s_3 d_4 \\ d_3 \end{pmatrix}$.

The vector between from the shoulder and the elbow in the shoulder-aligned frame is $a = \begin{pmatrix} a_2 \\ d_2 \\ 0 \end{pmatrix}$

By additivity, we can now express the vector from the shoulder to the wrist center in the shoulder-aligned frame as $a + b = \begin{pmatrix} a_2 + c_3 d_4 \\ d_2 + s_3 d_4 \\ d_3 \end{pmatrix}$. To express it in the waist-aligned frame, we simply remove the shoulder angle q_2 by multiplying by $R_y(q_2)$ on the left. Note that is vector is also $[X, Y, Z]$:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = R_y(q_2)(a + b)$$

We can sum the square components to obtain $N = X^2 + Y^2 + Z^2$:

$$N = (a_2 + c_3 d_4)^2 + (d_2 + s_3 d_4)^2 + (d_3)^2 = a_2^2 + 2a_2 d_4 c_3 + d_4^2 c_3^2 + d_2^2 + 2d_2 d_4 s_3 + d_4^2 s_3^2 + d_3^2 = a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 s_3$$

Knowing that $Y = d_2 + d_4 s_3 \implies s_3 = \frac{Y - d_2}{d_4}$, we can replace s_3 in the previous expression and then isolate c_3:

$$N = a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 \frac{Y - d_2}{d_4} = a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 Y - 2d_2^2$$

Let $K = d_2^2 - a_2^2 - d_3^2 - d_4^2$

$$\implies c_3 = \frac{N + K - 2d_2 Y}{2a_2 d_4}$$

and thus $q_3 = \text{atan2}(s_3, c_3)$.

To find q2:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = R_y(q_2)$$

To solve q1 we will construct a degree-4 polynomial.

First note that $X^2 + Y^2 + Z^2 = ||R_x\left(-\frac{\pi}{6}\right) v|| = ||v||$.

$$\implies X^2 + Y^2 + Z^2 = (c_1 x_w + s_1 y_w - a_1)^2 + (-s_1 x_w + c_1 y_w)^2 + (z_w - d_1)^2$$

$$= (c_1 x_w + s_1 y_w)^2 - 2c_1 x_w a_1 - 2s_1 y_w a_1 + a_1^2 + s_1^2 x_w^2 - 2s_1 c_1 x_w y_w w + c_1^2 y_w^2 + z_w^2 - 2z_w d_1 + d_1^2$$
$$= c_1^2 x_w^2 + 2c_1 s_1 x_w y_w + s_1^2 y_w^2 - 2c_1 x_w a_1 - 2s_1 y_w a_1 + a_1^2 + s_1^2 x_w^2 - 2s_1 c_1 x_w y_w w + c_1^2 y_w^2 + z_w^2 - 2z_w d_1 + d_1^2$$
$$= x_w^2 + y_w^2 - 2x_w c_1 a_1 - 2y_w s_1 a_1 + a_1^2 + z_w^2 - 2z_w d_1 + d_1^2$$

which happens to be a linear function of (c_1, s_1).

From the past expressions of s_3 and c_3 found when solving for q_3, and noting that $c_3^2 + s_3^2 = 1$, we have

$$4a_2^2(Y - d_2)^2 + (N + K - 2d_2 Y)^2 = 4a_2^2 d_4^2$$

using the half angle formulas

$$t = \tan(\frac{\theta}{2}),\ \cos(theta) = \frac{1 - t^2}{1 + t^2},\ \sin(theta) = \frac{2t}{1 + t^2}$$